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Productized & Enterprise RAG

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Speaker

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- Head of Engineering, Romania
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- Mentor @Google Developer Groups Romania
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- Supervised 12 BSc and MSc diploma projects

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Agenda

01 Existing repos & products

02 Pros, Cons, Limitations and Considerations

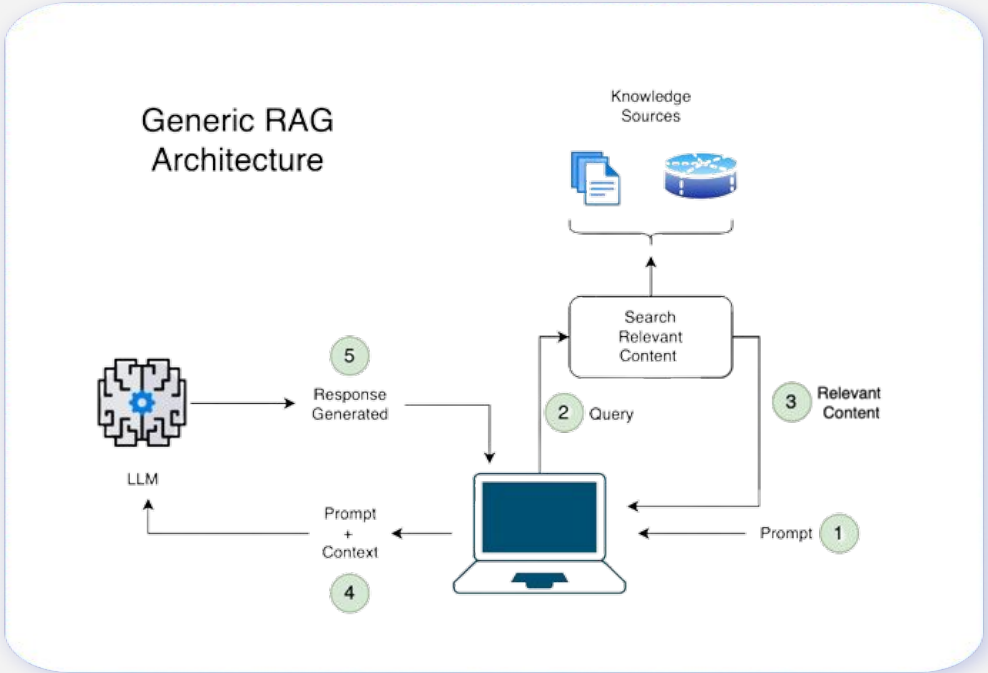
Quick recap

What is RAG

A RAG system essentially correlates a **user's prompt** with a relevant **data chunk.** It does this by identifying **the most semantically similar** chunk from the database.

This chunk then becomes **the context** for the prompt.

When **passed to the Large Language Model (LLM),** it enables the system to provide a relevant answer within the given context.



Why do we need RAG?

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- **Expands Knowledge Base**
  RAG accesses a vast external database, enriching its knowledge beyond initial training data
- **Improves Accuracy**
  Enhances response precision by integrating relevant, real-time information
- **Adaptable**
  Effectively handles novel and niche queries
- **Increases Efficiency**
  Streamlines information retrieval and generation process
- **Versatile Applications**
  Useful across various fields, from customer support to research

Source: What Is Retrieval-Augmented Generation, aka RAG?

LLMs are limited to the knowledge they were originally trained on, and retraining or fine-tuning them is slow and expensive.

Embeddings. Similarity

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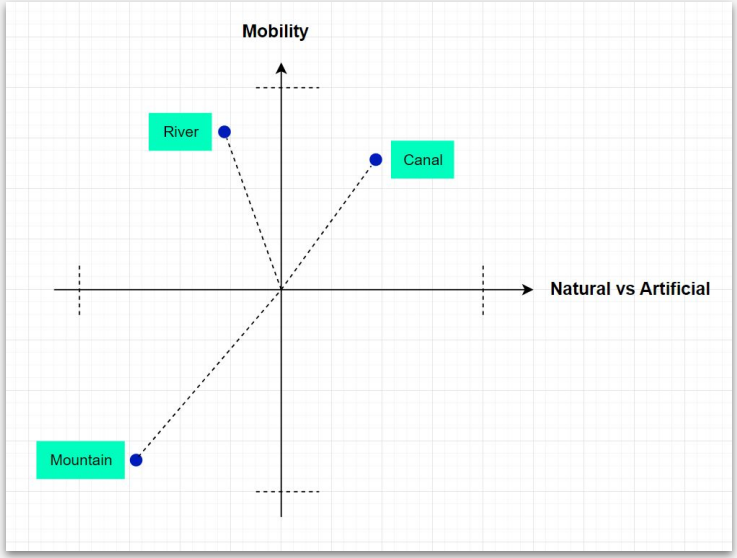
- **Embeddings**
  Numerical representations of concepts, in a high-dimensional space, capturing semantic meaning.

- **Similarity**:
  - Lexical: entities are alike in appearance
  - **Semantic**: entities are alike in meaning

- In RAG we represent entities **by describing** them.
  This is a form of **knowledge representation**.



Read more: Wikipedia - Cosine Similarity

Example: Mountain, River, Canal

One hot encoding

Mountain:	1
River:	2
Canal:	3

2-Dimensional Space

	[Natural vs Artificial, Mobility]
Mountain:	[-0.7, -0.8]
River:	[-0.3, 0.7]
Canal:	[0.4, 0.5]

Multilinguality

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Generated with DALL·E 3

- **Multilingual Capabilities**
  RAG and chatbots can process and respond in multiple languages, enhancing global reach and accessibility.
- **Cross-lingual Understanding**
  These systems understand queries in one language and retrieve information in another, facilitating cross-lingual interactions.
- **Language-Agnostic Embeddings**
  Use of embeddings that capture semantic meanings across languages.
- **Challenges**
  Handling idiomatic expressions, cultural nuances, and varying sentence structures in different languages.

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Existing repos & products

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Custom GPTs

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OpenAI

November 6, 2023 Product

Introducing GPTs

You can now create custom versions of ChatGPT that combine instructions, extra knowledge, and any combination of skills.

LM Studio

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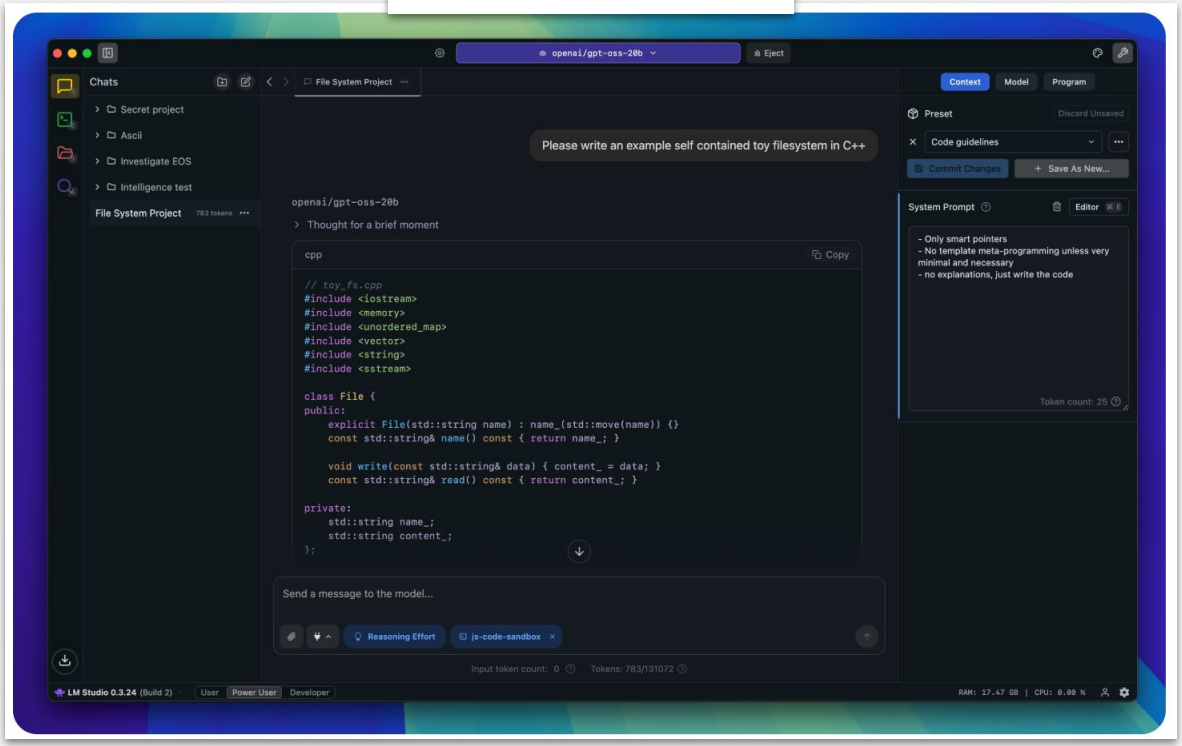
LM Studio



Anything LLM

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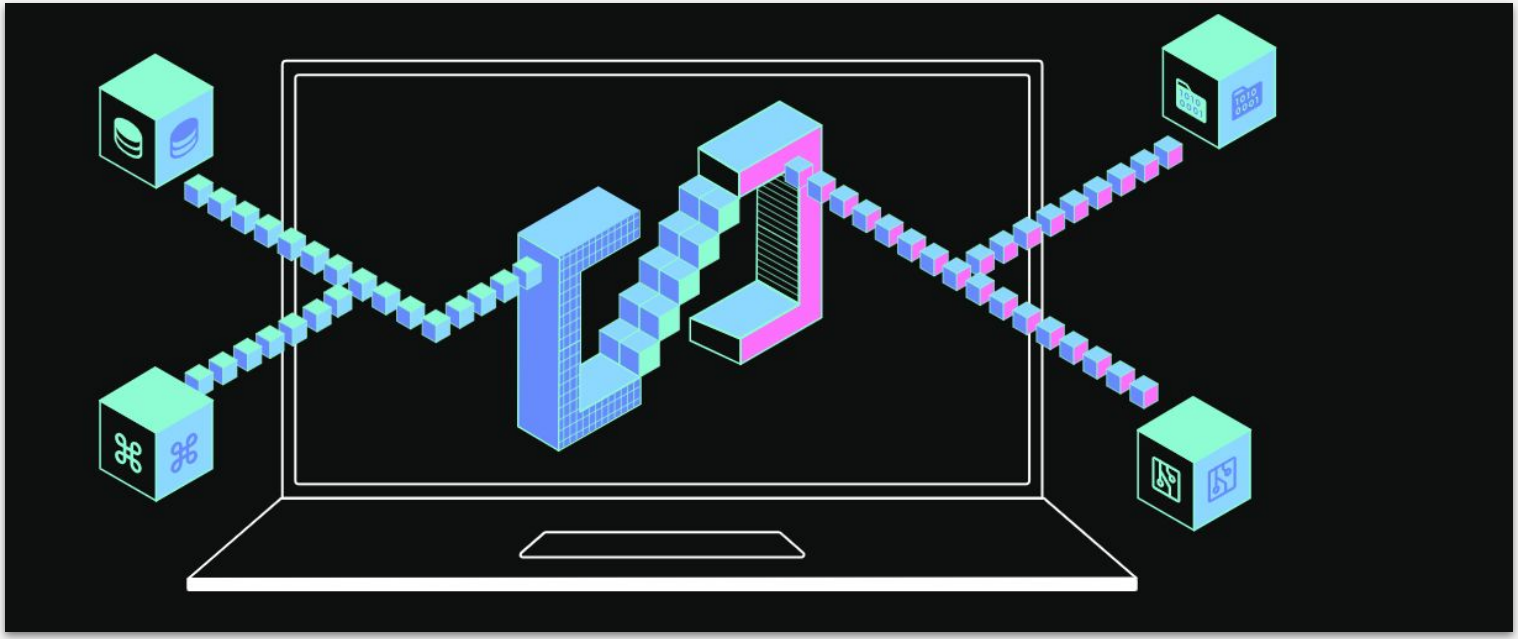
Anything LLM



OpenWeb UI

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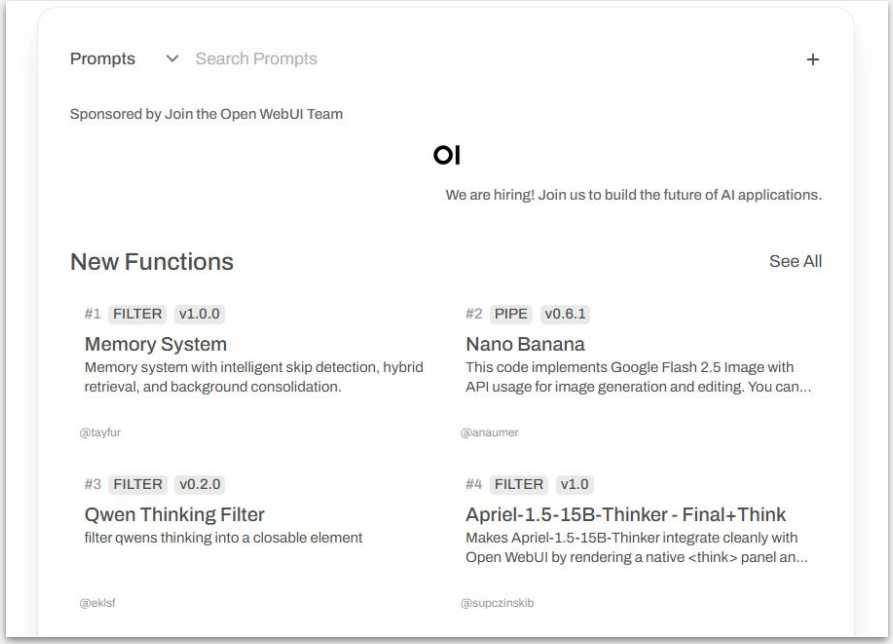
Open WebUI



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Glean Assistant

Can you do a 360 on Acme Corp?

- Searching company knowledge
- Pulling account activity from Salesforce
- Surfacing recent notes

Glean Search

Sourcegraph

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Sourcegraph

context:global Search for code or files...

Aa

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Search in files, paths, and repository-names

- `test server` (both terms anywhere)
- `"Error 1001"` (specific string)
- `"\"Error 1001\""` (specific string containing quotes)
- `foo OR bar`
- `/open(File|Dir)/` (regular expression)

Search in commit diffs

- `type:diff after:1week fix`
- `type:diff author:alice add`

Filter by...

- `file:file.go foo`
- `repo:org/repo`
- `repo:org/` (all repositories in org)
- `lang:javascript`

Advanced

- `repo:has.description(foo)`
- `repo:has.commit.after(1week)`
- `file:^some_path file:has.owner(alice)`
- `file:^some_path select:file.owners`

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Pros, Cons, Limitations and Considerations

Pros & Cons

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✅ Pros

- Enhances factual accuracy via external knowledge
- Reduces hallucinations vs. pure LLMs
- Enables dynamic updates without retraining
- Modular, domain-adaptable architecture
- Scalable across multimodal or multi-source data

⚠️ Cons

- Retrieval quality depends on data prep & embeddings
- Latency increases with retrieval pipeline
- Requires infra (vector DBs, indexing, chunking, etc.)
- Context window still limits answer depth

Limitations & Considerations

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🚧 Limitations

- Not reasoning-first, still retrieval-driven
- Poor handling of implicit or fuzzy queries
- No built-in memory or learning from feedback
- Struggles with conflicting or outdated sources

💡 Considerations

- Balance retrieval vs. generation cost & latency
- Continuous evaluation of relevance & drift
- Governance for data freshness and accuracy
- Potential upgrade path: **Agentic / Tool-use RAG**

Enterprise Data Protection and Privacy

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- **Safeguarding sensitive information** and maintaining customer trust.
- **Regulatory Compliance**: Adherence to laws like GDPR, which mandate data security and privacy.
- **Encryption & Anonymization**: Secure data and protect individual identities.
- **Access Control**: Implementing strict permissions and access policies to sensitive data.
- **Regular Audits**: Monitoring and auditing to identify and rectify any vulnerabilities.
- Importance of using **External Data** & RAG models.
- **Augment context** in order to get accurate completions.

Image by Jan Alexander from Pixabay

Quiz Time

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Q1

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What is the main goal of RAG?

A. To train larger models

B. To reduce hallucinations using external knowledge

C. To replace fine-tuning

D. To speed up token generation

Q2

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What is a key limitation of RAG systems?

A. They cannot retrieve multimodal data

B. They require GPU inference for retrieval

C. They depend heavily on data quality and embeddings

D. They cannot use APIs

Q3

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What's the next evolution beyond basic RAG?

A. Fine-tuned GPT models

B. Static FAQ chatbots

C. Agentic or tool-using RAG architectures

D. Prompt templates with zero retrieval

Key Takeaways from the session

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- Boosts **factual accuracy** by grounding LLMs in **real data**
- **Reduces hallucinations**, but depends on retrieval quality
- **No retraining needed**, knowledge updated via data refresh
- Latency–complexity **trade-off**: retrieval adds overhead
- Still **limited** reasoning and context awareness
- Next **evolution** → Agentic, monitored, tool-using RAGs

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Thank you!